IGARSS 2023 - 2023 IEEE International Geoscience and Remote Sensing Symposium | 979-8-3503-2010-7/23/$31.00 ©2023 IEEE | DOI: 10.1109/IGARSS52108.2023.10281962

POTENTIAL OF DEEP LEARNING FOR FOREST HEIGHT ESTIMATION FROM TANDEM-X BISTATIC INSAR DATA

Daniel Carcereri[1,2], *Paola Rizzoli*[1], *Dino Ienco*[3], *and Lorenzo Bruzzone*[2]

[1]German Aerospace Center (DLR), Microwaves and Radar Institute, Germany
[2]Universitá degli studi di Trento (UNITN), Italy
[3]National Research Institute for Agriculture, Food and the Environment (INRAE), University of Montpellier, France

ABSTRACT

Large-scale and up-to-date canopy height model (CHM) estimates are key to forest resources assessment and disturbance analysis. In this work we present an investigation of the potential of Deep Learning (DL) for the regression of forest height from TanDEM-X bistatic InSAR data. We propose a novel fully convolutional neural network (CNN) framework, trained and tested on four tropical sites in Gabon, Africa, together with a series of experiments for assessing the impact of different input features with specific focus on bistatic InSAR. The obtained results are extremely promising and already in line with state-of-the-art methods based on theoretical modelling, with the remarkable advantage of requiring only one single TanDEM-X acquisition at inference time.

Index Terms— forest monitoring, forest height, deep learning, InSAR, TanDEM-X

1. INTRODUCTION

Forests are one of the most widespread ecosystems on the planet, estimated to cover 31% of Earth's total surface. The monitoring of these areas is typically performed by measuring proxy indices, such as the height of the canopy or the above-ground biomass. Measurements can be directly and accurately performed in-situ [1], but they are typically very expensive and time consuming, overall limiting the coverage and repeatability of these surveys [2]. Remote sensing (RS) systems offer access to medium-to-large scale products, which can be used to derive forest biophysical parameters from the acquired data. In particular, LiDAR systems are widely used in the creation of regional or country-scale canopy height maps (CHM). Airborne mounted sensors can typically sample forests very accurately, but posses limited spatial coverage and revisit capabilities, in contrast to spaceborne systems, which can achieve global coverage but are limited in resolution by the current technology [3]. Present spaceborne optical and radar imaging systems overcome these limitations, offering global coverage and revisit times in the order of days, with ample availability of freely accessible products. As a drawback, it is not possible to directly measure forest parameters from the acquired set of images, requiring instead extensive modelling in order to derive parameters estimates [4]. A common compromise in the generation of large-scale products is to use in-situ or LiDAR-derived measurements to fit the allometric relationships between forest parameters and RS observables [1]. The resulting models allow for the generation of large-scale estimates, but typically fail to capture the full complexity of the underlying relationships, limiting their application to the specific geographic locations and physical conditions seen during the calibration process [5]. Physical-based models can overcome some of these limitations, achieving a better accuracy and higher resolution estimates. Nevertheless, model initialization and parameter inference typically require auxiliary information [4] or large amounts of acquisitions, limiting the temporal frequency at which predictions can be produced.

Given the latest advancements in the field of computer vision and the large availability of open-access earth observation (EO) product, data-driven approaches have been constantly gaining attention in the context of RS applications [6]. In particular, deep learning (DL) techniques have been already applied with great success to classification problems, but remain largely unexplored for the estimation of biophysical parameters from RS data [6]. The aim of this work is therefore to investigate the use of DL for forest parameters regression and, in particular, to focus on the use of TanDEM-X interferometric SAR (InSAR) data. Our objective is to explore the potential of this unique datasource that, relying on the bistatic nature of TanDEM-X, allows for the acquisition of single-pass InSAR data unaffected by temporal decorrelation. We propose the definition of a novel DL framework capable of generating large-scale, high resolution and potentially frequently updatable canopy height estimates. To this end, we train our model to regress forest heights from TanDEM-X data in a supervised manner. This is accomplished by utilizing reference data derived from LiDAR canopy height maps acquired during the 2016 AfriSAR campaign. Then we present a detailed analysis for assessing the model perfor-

Authorized licensed use limited to: Turun Yliopisto. Downloaded on March 05,2026 at 10:55:14 UTC from IEEE Xplore. Restrictions apply.

mance and the added value of bistatic InSAR with respect to a classical SAR configuration.

2. METHODOLOGY

2.1. CNN Architecture

Since spatial context is required to improve the estimation accuracy [6], our regression framework consists of a deep convolutional neural network (CNN). The proposed model is fully convolutional (Fig. 1) and is composed of a sequence of functional blocks. The blocks consists of a two-dimensional spatial convolutional layer, a batch normalization layer and an activation function. The input block uses 1×1 kernel filters, to gradually increase the input feature dimensionality to 128. All 10 hidden blocks use 128 3×3 kernels. Finally, the output block gradually reduces the feature dimensionality to the single forest height estimate using 1×1 kernels. All blocks apply a rectified linear unit (ReLU) activation function, except from the regression head, which drops the batch normalization layer and applies a linear (i.e. pass-through) activation function. We used a supervised training approach on 15×15 pixel patches, chosen to guarantee that the model retains enough spatial context for proper estimates. The patches are sampled with overlap and without repetition from areas corresponding to forested pixels in the reference. As loss function, we used the mean square error (MSE) of the estimates, with the addition of the L2 norm of the networks weights as regularization term:

$$Loss = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2 + \lambda \cdot \sum_{j=1}^{m} w_j^2, \qquad (1)$$

where $\hat{y}_i$ is the predicted canopy height value, y_i is the corresponding reference value, w_j are the trainable model weights and $\lambda = 10^{-4}$ is the regularization parameter. Model weight initialization and training sample sequence are random, causing the stochastic gradient descent-based algorithm ADAM to converge to potentially different local minima. We thus trained five independent models, and selected the overall best model with respect to the validation data. For each experiment, we evaluated the selected model over an independent test-set, and quantified regression performance using the mean error (ME), the mean absolute error (MAE), the mean absolute percentage error (MAPE), the root mean-squared error (RMSE) and the coefficient of determination (R^2) of the estimates.

2.2. Dataset Description

To train, validate and test the proposed model, we considered the sites in Gabon (Africa) covered by the AfriSAR campaign (NASA/ESA) dataset. Each area was surveyed between February and March of 2016, using NASA's airborne-mounted LVIS LiDAR sensor [3]. Wall-to-wall CHM estimates with a ground sampling distance of 25m have been generated from airborne LiDAR data acquired between February and March of 2016 [3]. The location of the different test sites is depicted in Fig. 2.

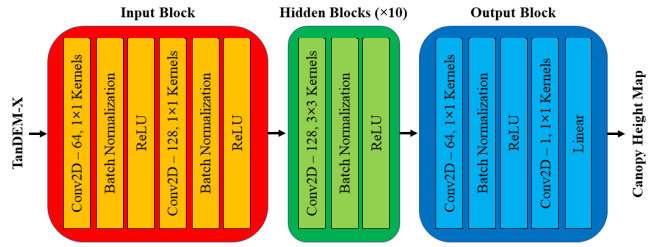

Fig. 1. Proposed deep CNN architecture for forest parameters regression.

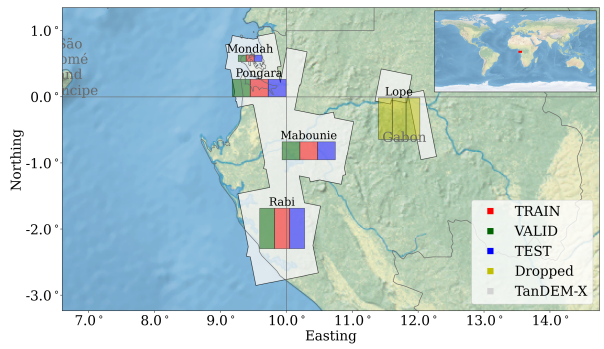

Fig. 2. The study-areas in Gabon (Africa) and their subdivisions into three subsets for training, validation and testing. The TanDEM-X footprints are identified in gray.

The main set of input features to the network consists of bistatic InSAR products acquired using DLR's TanDEM-X constellation. The TanDEM-X mission comprises two twin satellites orbiting in close formation and acting as single-pass interferometer at X band. The quality of the interferometric product is typically quantified through the *interferometric coherence* γ_{tot}, consisting of the the normalized cross-correlation between the input images pair. Crucially, the InSAR process is affected by multiple sources of noise, which are related to both the properties of the system and the characteristics of the illuminated targets [7]. The peculiarity of the TanDEM-X mission is the single-pass acquisition capability of InSAR data, which avoids the critical effects of temporal decorrelation. Moreover, the volume decorrelation factor γ_{vol}, quantifies the amount of decorrelation caused by volumetric scattering [7]. The effect becomes significant over vegetated areas, as the canopy acts like a superposition of vertically displaced scatterers. For this work, we selected all the suitable TanDEM-X acquisitions covering the AfriSAR

Authorized licensed use limited to: Turun Yliopisto. Downloaded on March 05,2026 at 10:55:14 UTC from IEEE Xplore. Restrictions apply.

study-sites, acquired within a period spanning from 2015 to 2016 and assuming changes in forest height to be negligible over such a time span. These constraints resulted in a low coverage of the Lopeé test-site, prompting us to remove it from the considered list of test areas. For the estimation of the total interferometric coherence γ_{tot}, we applied Φ-Net [8] and derived the volume decorrelation factor γ_{vol} by following the estimation procedure presented in [9]. Finally, for each area we generated a DEM starting from the global TanDEM-X edited DEM product. The resulting input feature stack to the network is composed of the backscattering coefficient σ^0 in HH polarization, the local incidence angle θ_{inc}, the total interferometric coherence γ_{tot}, the corresponding volume decorrelation factor γ_{vol}, the acquisition height of ambiguity h_{amb} and the DEM.

3. RESULTS

We evaluate the baseline regression performance of our framework by considering all input features described in Sec. 2.2. The resulting prediction accuracy metrics are presented in Tab. 1. Fig. 3 shows an overall good agreement between the predictions and the references, except for a cluster of heavily underestimated tall forest samples corresponding to a phase-unwrapping error in the DEM covering the Pongara National Park. The predicted CHM map over this area is presented in Fig. 4. The performance also appears to be correlated to the distribution of canopy heights for the different test-sites. The Mondah forest is predominantly characterized by large extents of low vegetation, with isolated patches of tall forest. This reflects on the MAE and MAPE metrics, which are the smallest and the largest among all test sites, respectively. With lower tree heights the absolute error scales accordingly, but the relative estimation error has on average a bigger impact. The sites of Mabounié and Rabi show similar behaviours, as they are predominantly covered by tall forest structures in the range of 25m to 45m of height. In this scenario the model uncertainty appears to increase, with the R^2 metric in particular being penalized by the very clustered concentration of tree heights. The MAE for both sites also worsens in this case, whereas the estimation error for the taller tree stand population has a lower impact on the MAPE. Notably, no performance metric is on its own sufficient to conclusively evaluate the overall prediction performance of the model. This is due to the different scenarios and, in particular, to their different canopy height composition that has different impacts on the accuracy metrics.

One of the main advantages of the TanDEM-X constellation over its alternatives is the capability to acquire bistatic InSAR information by design. To assess the relative importance of SAR and InSAR features, we alternatively drop γ_{tot}, γ_{vol} and h_{amb}, or σ^0 from the input sets. Repeating the aforementioned training and evaluation steps yields the results presented in Tab. 2 and Tab. 3, respectively. Both scenarios see a decrease in overall performance when compared to the baseline case. Notably, removing the backscatter information appears to have a significantly smaller impact than the removal of the InSAR features. In particular, results for the SAR-only scenario show a decrease of the MAPE of 2.124% and of RMSE of 0.675m compared to the InSAR scenario, and of

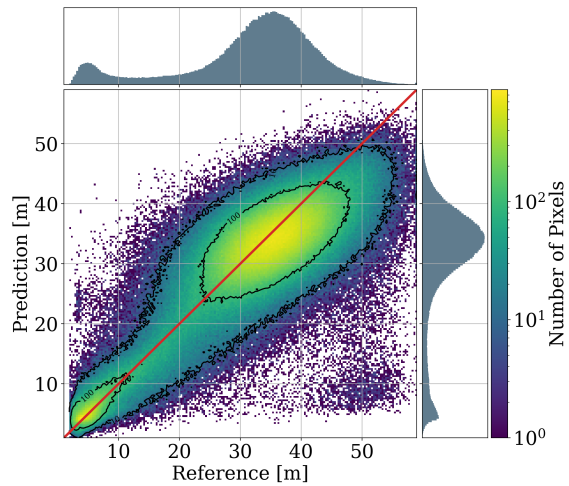

Fig. 3. Scatterplot in logarithmic scale of the predicted and reference canopy height values over all four test sites in Gabon, Africa, together with the side distributions of both the reference and the prediction.

Test-Sites	N° Acq.	ME	MAE	MAPE	RMSE	R^2	N° Pixels
Unit		[m]	[m]	[%]	[m]	[·]	
Mabounié	4	-0.753	4.410	14.215	5.735	0.460	410018
Mondah	4	0.308	2.488	27.513	3.645	0.885	88371
Pongara	2	-2.306	3.922	16.207	6.423	0.823	253394
Rabi	4	-1.784	4.394	13.602	5.602	0.404	714164
TOT	12	-1.459	4.201	15.062	5.692	0.734	1465947

Table 1. Test-site and overall prediction accuracies.

Test-Sites	N° Acq.	ME	MAE	MAPE	RMSE	R^2	N° Pixels
Unit		[m]	[m]	[%]	[m]	[·]	
Mabounié	4	-1.336	4.937	15.713	6.505	0.306	410018
Mondah	4	0.103	2.227	22.136	3.479	0.895	88371
Pongara	2	-2.344	3.831	16.558	6.277	0.831	253394
Rabi	4	-1.511	4.503	13.943	5.738	0.374	714164
TOT	12	-1.508	4.371	15.383	5.949	0.709	1465947

Table 2. Test-site and overall prediction accuracies without the backscatter information.

Test-Sites	N° Acq.	ME	MAE	MAPE	RMSE	R^2	N° Pixels
Unit		[m]	[m]	[%]	[m]	[·]	
Mabounié	4	-0.351	4.810	15.845	6.250	0.359	410018
Mondah	4	0.140	2.585	26.836	3.906	0.868	88371
Pongara	2	-2.742	4.664	21.558	7.446	0.763	253394
Rabi	4	-3.159	5.288	15.871	6.792	0.123	714164
TOT	12	-2.102	4.883	17.507	6.624	0.639	1465947

Table 3. Test-site and overall prediction accuracies without the bistatic InSAR information.

Authorized licensed use limited to: Turun Yliopisto. Downloaded on March 05,2026 at 10:55:14 UTC from IEEE Xplore. Restrictions apply.

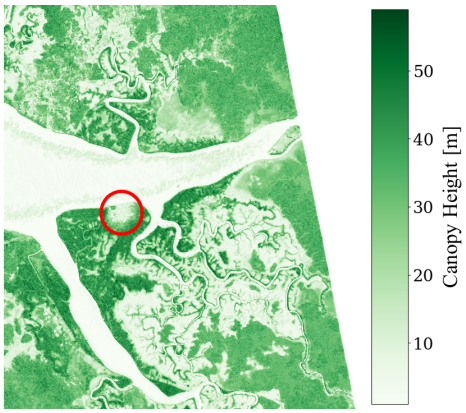

Fig. 4. Inference example over the Pongara National Park. The red circle shows an area of heavy underestimation due to a phase unwrapping error in the global TanDEM-X DEM.

2.445% and 0.932m, respectively, compared to the baseline scenario.

4. CONCLUSIONS AND OUTLOOK

In this work, we have proposed a framework based on a deep learning model for the generation of canopy height estimates from a single TanDEM-X bistatic InSAR product. The definition of our framework targets the generation of operational estimates at country-level scale in accordance with the availability of satellite imagery. We trained and tested our model across four different sites in the tropical region of Gabon. The evaluation metrics underlined a good agreement between the predictions and the reference data, demonstrating the validity of bistatic interferometric X-band SAR data for the estimation of forest heights. Future works will concentrate on the generalization of the DL framework at large-scale and on the exploitation of the implicit multi-modal properties of the proposed CNN to perform data fusion with other spaceborne sensors.

5. REFERENCES

[1] T. Jucker, J. Caspersen, J. Chave, C. Antin, N. Barbier, F. Bongers, M. Dalponte, K. Y. van Ewijk, D. I. Forrester, M. Haeni, S. I. Higgins, R. J. Holdaway, Y. Iida, C. Lorimer, P. L. Marshall, S. Momo, G. R. Moncrieff, P. Ploton, L. Poorter, K. Abd Rahman, M. Schlund, B. Sonké, F. J. Sterck, A. T. Trugman, V. A. Usoltsev, M. C. Vanderwel, P. Waldner, B. M. M. Wedeux, C. Wirth, H. Wöll, M. Woods, W. Xiang, N. E. Zimmermann, and D. A. Coomes, "Allometric equations for integrating remote sensing imagery into forest monitoring programmes," *Global Change Biology*, vol. 23, no. 1, pp. 177–190, 2017.

[2] N. Picard, L. Saint-André, and M. Henry, *Manual for building tree volume and biomass allometric equations from filed measurement to prediction*, Food and Agriculture Organization of the United Nations (FA0), 2012, OCLC: 931325352.

[3] T. Fatoyinbo, J. Armston, M. Simard, S. Saatchi, M. Denbina, M. Lavalle, M. Hofton, H. Tang, S. Marselis, N. Pinto, S. Hancock, B. Hawkins, L. Duncanson, B. Blair, C. Hansen, Y. Lou, R. Dubayah, S. Hensley, C. Silva, J. R. Poulsen, N. Labrière, N. Barbier, K. Jeffery, D. Kenfack, M. Herve, P. Bissiengou, A. Alonso, G. Moussavou, L. T.J. White, S. Lewis, and K. Hibbard, "The NASA AfriSAR campaign: Airborne SAR and lidar measurements of tropical forest structure and biomass in support of current and future space missions," *Remote Sensing of Environment*, vol. 264, pp. 112533, 2021.

[4] F. Kugler, D. Schulze, I. Hajnsek, H. Pretzsch, and K. P. Papathanassiou, "TanDEM-X Pol-InSAR Performance for Forest Height Estimation," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 52, no. 10, pp. 6404–6422, Oct. 2014.

[5] C. Torres de Almeida, L. S. Galvão, L. E. d. O. C. e Aragão, J. P. H. B. Ometto, A. D. Jacon, F. R. d. S. Pereira, L. Y. Sato, A. P. Lopes, P. M. L. d. A. Graça, C. V. d. J. Silva, J. Ferreira-Ferreira, and M. Longo, "Combining LiDAR and hyperspectral data for aboveground biomass modeling in the brazilian amazon using different regression algorithms," *Remote Sensing of Environment*, vol. 232, pp. 111323, 2019.

[6] N. Lang, K. Schindler, and J. D. Wegner, "Country-wide high-resolution vegetation height mapping with sentinel-2," *Remote Sensing of Environment*, vol. 233, pp. 111347, 2019.

[7] M. Martone, P. Rizzoli, and G. Krieger, "Volume decorrelation effects in TanDEM-x interferometric SAR data," *IEEE Geoscience and Remote Sensing Letters*, vol. 13, no. 12, pp. 1812–1816, 2016.

[8] F. Sica, G. Gobbi, P. Rizzoli, and L. Bruzzone, "ϕ-net: Deep residual learning for insar parameters estimation," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 59, no. 5, pp. 3917–3941, May 2021, Code available at: https://github.com/DLRRadarScienceGroup/Phi-Net.

[9] P. Rizzoli, L. Dell'Amore, J.L. Bueso-Bello, N. Gollin, D. Carcereri, and M. Martone, "On the derivation of volume decorrelation from Tand-DEM-X bistatic coherence," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 15, pp. 3504–3518, Apr 2022.

Authorized licensed use limited to: Turun Yliopisto. Downloaded on March 05,2026 at 10:55:14 UTC from IEEE Xplore. Restrictions apply.